

$^{35}\text{Cl}(\text{p},\text{p}'\gamma)$ 1969Du08,1970Ta02,1972Br45

Also include (p,p') measurements listed below.

$^{35}\text{Cl}(\text{p},\text{p}'\gamma)$ measurements:

1969Du08: E=4.6-6.2 MeV 0.05 μA proton beam. Thin targets of CdCl_2 (99% ^{35}Cl), 100 $\mu\text{g}/\text{cm}^2$. A 300 μm annular surface-barrier detector on the beam axis and a movable 390 μm surface-barrier detector rotated from 20° to 148° relative to beam direction for detecting protons, FWHM=24 keV; a 7.62 by 7.62-cm-diam $\text{NaI}(\text{Tl})$ crystal and 15- and 35-cc $\text{Ge}(\text{Li})$ for detecting γ -rays. Measured $\sigma(\text{E}_\text{p},\theta)$, E_γ , I_γ , $\text{p}\gamma(\theta)$. Deduced levels, J^π , branchings, mixing ratios, half-lives using Doppler Shift Attenuation Method (DSAM).

1970Ta02: E=4.24, 4.82, 5.08 and 5.11 MeV proton beams produced from the Universite Montreal EN tandem accelerator. Targets of 16 $\mu\text{g}/\text{cm}^2$ AgCl evaporated onto 0.013 cm gold foils, 99% enriched in ^{35}Cl . A 12.70-cm-diam by 15.24-cm-long $\text{NaI}(\text{Tl})$ and a 30 cm^3 $\text{Ge}(\text{Li})$ detector for detecting γ -rays. Measured E_γ , I_γ , $\gamma(\theta)$, $\gamma(\text{lin pol})$. Deduced levels, J^π , branchings, mixing ratios.

1972Br45: E=3.45-6.00 MeV protons produced from the University of Auckland folded tandem accelerator. Natural chlorine targets of AgCl , PbCl_2 and solid Cl (75.77% ^{35}Cl , 24.33% ^{37}Cl). A 25 cm^3 $\text{Ge}(\text{Li})$ detector for detecting γ -rays, FWHM=2.3 keV for $\text{E}_\gamma=1.332$ MeV. Measured E_γ . Deduced levels, half-lives using DSAM. **1972Br45** report a few mixing ratios in their TABLE V, but it is unclear where they are taken from. They are definitely not from **1972Br45** since this work is only focused on lifetime measurements.

1972Va06: E=5.47-5.61 MeV protons from the 5.5 MV Van de Graaff accelerator of the Southern Universities Nuclear Institute. Targets of BaCl_2 (99% ^{35}Cl) on gold backings. A 57 cm^3 $\text{Ge}(\text{Li})$ detector. Measured E_γ . Deduced half-lives using DSAM.

1961St09: E=2.3-3.25 MeV. $\text{NaI}(\text{Tl})$ detector. Targets of AgCl . Measured $\sigma(\text{E}_\text{p},\theta)$, $\gamma(\theta)$. Deduced levels, J for the levels of 1.22 and 1.76 MeV.

1966Er06: E=0.733-2.595 MeV protons produced from the Utrecht 3 MV Van de Graaff accelerator. BaCl_2 target (99.5% ^{35}Cl) on a tantalum backing. Cylindrical NaI assemblies, 10-cm by 10-cm. Measured E_γ , $\gamma(\theta)$. Deduced levels, J^π , mixing ratios, mainly for ^{36}Ar and for the ^{35}Cl level at $\text{E}_\text{x}=1760$.

1970Ta09: E=5.11 MeV proton beam. $\text{NaI}(\text{Tl})$ detector. Measured E_γ . Developed an analytical method to interpret the linear polarization measurement of two unresolved γ -rays: 3.00 and 3.16 MeV. Same authors as **1970Ta02**.

1971Ca40: E=4.15-6.15 MeV protons from the Oxford University Tandem accelerator. Targets of 20 $\mu\text{g}/\text{cm}^2$ natural BaCl_2 on gold, lead, and carbon backings. A 32 cm^3 $\text{Ge}(\text{Li})$ detector. Measured E_γ . Deduced half-lives for the levels of 2646 and 2695 keV using DSAM.

1974Hu09: E=5.65 MeV protons from the pulsed beam facility of the University of Alberta Van de Graaff accelerator. Targets of natural KCl on tantalum backing and PbCl_2 on gold backing. A 24 cm^3 coaxial $\text{Ge}(\text{Li})$ detector. Measured E_γ . Deduced half-life for the level of 3162 keV.

Other (p,p' γ) measurement: **1985Ki07** and **1975De16** (measured 1219 γ yield).

(p,p') measurements:

1969Ta13,1971Ta02: E=3.115 and 3.345 MeV protons produced from the Universite de Montreal Model EN tandem accelerator. AgCl target (99% ^{35}Cl) of 16 $\mu\text{g}/\text{cm}^2$ on gold backing. A 12.70-cm-diam by 15.24-cm-long $\text{NaI}(\text{Tl})$ for detecting γ -rays. Measured E_γ , I_γ , $\gamma(\theta)$. Deduced levels, J^π and mixing ratio for the level of 1763 keV. Same authors as **1970Ta02**.

1969An35: E=6.03 MeV proton beams on a CdCl_2 target. Inelastic protons detected by a $\text{Si}(\text{Li})$ detector. Measured $\sigma(\text{E}_\text{p},\theta)$. Deduced J for a level at 3930.

Other (p,p') measurement: **1993Bo40** (measured σ).

 ^{35}Cl Levels

$\text{E}(\text{level})^\dagger$	$J^\pi^\ddagger$	$\text{T}_{1/2}$	Comments
0	$3/2^+$		
1219.42 10	$1/2^+$	0.150 ps +42–28	J^π : $\text{p}\gamma(\theta)$ in 1969Du08 is consistent with spin=1/2. $\text{T}_{1/2}$: from $\tau=0.216$ ps +60–40, weighted average of 0.215 ps 75 (1972Va06), 0.21 ps +6–4 (1972Br45), 0.27 ps +19–10 (1969Du08).
1763.26 10	$5/2^+$	0.43 ps 20	J^π : $5/2^+$ from $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in 1966Er06 ; spin=5/2 also from $\text{p}\gamma(\theta)$ in 1969Du08 . $\text{T}_{1/2}$: from $\tau=0.62$ ps 28, unweighted average of 0.205 ps 75 (1972Va06), 0.49 ps +9–6 (1972Br45), 1.15 ps +45–63 (1969Du08).
2645.74 17	$7/2^+$	0.165 ps 35	J^π : $5/2^+$ from $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in 1970Ta02 ; spin $\geq 5/2$ from $\text{p}\gamma(\theta)$ in 1969Du08 . $\text{T}_{1/2}$: from $\tau=0.238$ ps 50, weighted average of 0.185 ps 50 (1972Va06) and 0.26 ps 6 (1972Br45), and 0.35 ps +10–7 (1969Du08 , using 2646 γ). Other: $\tau=0.29$ ps +37–13

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$^{35}\text{Cl}(\text{p},\text{p}'\gamma)$ **1969Du08,1970Ta02,1972Br45 (continued)** ^{35}Cl Levels (continued)

E(level) [†]	J ^π [‡]	T _{1/2}	Comments
2694.50 28	3/2 ⁺	43 fs 11	using 882γ also reported in 1969Du08, but is considered less reliable by authors; 0.083 ps 21 from 1971Ca40 is considered an outlier. J ^π : 3/2 ⁺ from γ(θ) and γ(lin pol) in 1970Ta02. T _{1/2} : from τ=63 fs 16, weighted average of 65 fs 20 (1972Va06) and 62 fs 16 (1969Du08, using 2695γ). Other: τ=115 fs +95–59 using 931γ also reported in 1969Du08, but is considered less reliable by authors; <120 fs (1972Br45); 11 fs 7 from 1971Ca40 is considered an outlier.
3002.6 5	5/2 ⁺	19 fs 9	J ^π : 5/2 ⁺ from γ(θ) and γ(lin pol) in 1970Ta02. T _{1/2} : from τ=27 fs 13, weighted average of 18 fs +26–15 (1972Va06) and 31 fs 13 (1969Du08). Other: <80 fs (1972Br45).
3162.75 10	7/2 [–]	32 ps 4	J ^π : spin=7/2 from γ(θ) and π=– from γ(lin pol) in 1970Ta02. T _{1/2} : from τ=46 ps 6 in 1974Hu09. Other: τ>17 ps (1972Va06), >10 ps (1972Br45).
3930 30	3/2		E(level),J ^π : from 1969An35. Spin is determined based on J-dependence of measured proton inelastic scattering cross sections and known assignments of 3163 and 3000 levels.

[†] From a least-squares fit to γ-ray energies with uncertainties, unless otherwise noted.[‡] From Adopted Levels. Supporting arguments from this dataset are given under comments if available. $\gamma(^{35}\text{Cl})$

E _i (level)	J _i ^π	E _γ [†]	I _γ [‡]	E _f	J _f ^π	Mult. [#]	δ [#]	Comments
1219.42	1/2 ⁺	1219.4 1	100	0	3/2 ⁺			E _γ : weighted average of 1219.4 1 (1972Br45) and 1219.3 3 (1969Du08). Γ _γ =27.43×10 ^{–4} eV (1969Du08). A ₂ =–0.025 29 for E _p =3.115 MeV; +0.007 8 for E _p =3.345 MeV (1969Ta13). A ₂ =+0.031 16, A ₄ =+0.014 20; A ₂ =–0.012 22, A ₄ =–0.003 30; A ₂ =–0.009 14, A ₄ =+0.014 6, for different beam energies (1961St09). A ₂ =+0.02 6, A ₄ =+0.05 9 (1969Du08).
1763.26	5/2 ⁺	1763.2 1	100	0	3/2 ⁺	M1+E2	–2.66 12	E _γ : weighted average of 1763.2 1 (1972Br45), 1763.3 3 (1969Du08). δ: weighted average of –3.0 5 from γ(θ,pol) in 1966Er06 and –2.64 12 from γ(θ,pol) in 1969Ta13 and 1971Ta02. Others: δ<–0.4 or δ>+0.4 from pγ(θ) in 1969Du08. A ₂ =–0.369 9, A ₄ =+0.213 13; A ₂ =–0.355 14, A ₄ =+0.200 20, for different beam energies (1969Ta13). A ₂ =–0.02 2, A ₄ =+0.07 2 (1969Du08). A ₂ =–0.141 13, A ₄ =+0.107 16; A ₂ =–0.225 29, A ₄ =+0.083 35; A ₂ =–0.049 12, A ₄ =–0.036 14, for different beam energies (1961St09).
2645.74	7/2 ⁺	882.3 [@] 3	13 5	1763.26	5/2 ⁺	M1+E2	+0.17 5	I _γ : unweighted average of 8.7 22 from 1970Ta02 and 17.6 35 from 1969Du08. %Branching=8 2 (1970Ta02), 15 3 (1969Du08). A ₂ =–0.372 34, A ₄ =+0.034 38; A ₂ =–0.533 31, A ₄ =+0.041 34, for different beam energies (1970Ta02). %Branching<1 (1970Ta02).
		1426 ^{&a} 2645.7 2	<1 ^{&} 100 2	1219.42	1/2 ⁺ 3/2 ⁺	E2		E _γ : weighted average of 2645.7 2 (1972Br45) and 2645.6 5 (1969Du08).

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$^{35}\text{Cl}(\text{p},\text{p}'\gamma)$ **1969Du08,1970Ta02,1972Br45 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. #	$\delta^\#$	Comments
								δ : $\delta(\text{O}/\text{Q})=0$ from $\gamma(\theta, \text{pol})$ in 1970Ta02 and $\text{p}\gamma(\theta)$ in 1969Du08. I_γ : from 1970Ta02. Other: 100 4 from 1969Du08. %Branching=92 2 (1970Ta02), 85 3 (1969Du08). $A_2=+0.22$ 5, $A_4=-0.13$ 6 (1969Du08). $A_2=+0.316$ 22, $A_4=+0.024$ 28; $A_2=+0.233$ 11, $A_4=+0.044$ 10; $A_2=+0.312$ 32, $A_4=-0.117$ 41; $A_2=+0.205$ 18, $A_4=-0.044$ 25, for different beam energies (1970Ta02).
2694.50	3/2 ⁺	931.3@ 3	18.6 25	1763.26	5/2 ⁺	M1+E2	+0.09 3	I_γ : weighted average of 19.0 25 (1970Ta02) and 17.7 38 (1969Du08). %Branching=15 2 (1970Ta02), 14 3 (1969Du08). $A_2=+0.018$ 46 (1970Ta02).
		1475@ 2	8.3 25	1219.42	1/2 ⁺	M1+E2	-0.63 21	I_γ : weighted average of 8.0 25 (1970Ta02) and 8.9 38 (1969Du08). %Branching=6 2 (1970Ta02), 7 3 (1969Du08). $A_2=-0.111$ 45 (1970Ta02).
		2694.1 6	100 3	0	3/2 ⁺	M1+E2	+0.26 2	E_γ : unweighted average of 2693.5 1 (1972Br45), 2694.6 5 (1969Du08). I_γ : from 1970Ta02. Other: 100 4 from 1969Du08. %Branching=79 2 (1970Ta02), 79 3 (1969Du08).
3002.6	5/2 ⁺	1239& 1	2.2& 11	1763.26	5/2 ⁺			I_γ : from 1970Ta02. %Branching=2 1 (1970Ta02).
		1783	3.3 11	1219.42	1/2 ⁺			E_γ : from level-energy difference. I_γ : weighted average of 5.4 32 (1970Ta02) and 3.1 11 (1969Du08). %Branching=5 3 (1970Ta02), 3 1 (1969Du08).
		3002.5 5	100 1	0	3/2 ⁺	M1+E2	+0.07 2	E_γ : weighted average of 3002.5 5 (1972Br45) and 3002.4 10 (1969Du08). I_γ : from 1969Du08. Other: 100 2 from 1970Ta02. δ : others: $-20 < \delta < -0.3$ or $+0.2 < \delta < +5$ (1969Du08, $\text{p}\gamma(\theta)$); $-0.02 < \delta < 0.09$ (1970Ta09, $\gamma(\text{lin pol})$). %Branching=93 2 (1970Ta02), 97 1 (1969Du08). $A_2=-0.30$ 4 (1969Du08). $A_2=-0.166$ 7, $A_4=+0.004$ 10; $A_2=-0.104$ 23, $A_4=+0.050$ 40; $A_2=-0.329$ 13, $A_4=-0.001$ 28; $A_2=+0.342$ 24, $A_4=+0.022$ 43, for different beam energies (1970Ta02).
3162.75	7/2 ⁻	517.0@ 8	19 8	2645.74	7/2 ⁺			I_γ : unweighted average of 11.1 22 (1970Ta02) and 26.6 63 (1969Du08). %Branching=10 2 (1970Ta02), 21 5 (1969Du08).
		1399&a	<1&	1763.26	5/2 ⁺			%Branching<1 (1970Ta02).
		1943&a	<1&	1219.42	1/2 ⁺			%Branching<1 (1970Ta02).

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$^{35}\text{Cl}(\text{p},\text{p}'\gamma)$ **1969Du08,1970Ta02,1972Br45** (continued) $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	Comments
3162.75	3162.6 1	100 2	0	3/2 ⁺	M2+E3	-0.22 5	E_γ : weighed average of 3162.6 1 (1972Br45) and 3162.5 10 (1969Du08). I_γ : from 1970Ta02. Other: 100 6 from 1969Du08. δ : weighted average of -0.14 +8-6 (1969Du08, $\text{p}\gamma(\theta)$) and -0.25 4 (1970Ta02, $\gamma(\theta,\text{pol})$). Other: -0.31< δ <-0.21 from $\gamma(\text{lin pol})$ in 1970Ta09. %Branching=90 2 (1970Ta02), 79 5 (1969Du08). $A_2=+0.48$ 21, $A_4=-0.13$ 73 (1969Du08). $A_2=+0.281$ 68, $A_4=+0.241$ 82; $A_2=+0.453$ 44, $A_4=+0.136$ 63; $A_2=+0.475$ 24, $A_4=+0.096$ 36, for different beam energies (1970Ta02).

[†] E_γ values with uncertainties for the ground-state transitions are not explicitly listed in 1972Br45 and 1969Du08 and the evaluators have taken their listed $E(\text{level})$ values and uncertainties (which are deduced from E_γ values by authors) as E_γ values of the corresponding ground-state transitions, considering ground-state transitions are dominant in each level.

[‡] Relative photon branching ratios deduced by the evaluators based on percentage branching ratios reported in 1970Ta02 and 1969Du08, which are given under comments where available.

[#] From $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in 1970Ta02, unless otherwise noted.

@ From 1969Du08.

& γ reported in 1970Ta02. Energy is from level-energy difference and intensity is from 1970Ta02. The ones with I_γ reported as upper limit are considered questionable by the evaluators.

^a Placement of transition in the level scheme is uncertain.

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Legend

Level Scheme

Intensities: Relative photon branching from each level

-----► γ Decay (Uncertain)